

Behavior "Lee 3"

1. **Puberty**: occurs from 6 to 8 months, while sexual maturity at 15-18 months for foreign breeds (360 kg) and 20-24 months for native heifers (260 kg)

البالوغ يبدأ من ٦ إلى ٨ أشهر لكن السيكسوال إلى عند الحيوان رقم ٢٠-٢٤ أشهر
 (أو) عند الوصول ٣٦٠ كيلو سلات أمبير (أو) ٢٤٠-٢٦٠ كيلو سلات أمبير

2. **Breeding Season**: Continuously cycling breeder shows peak fertility from "May to July" & low from "December to February" (2) (12) (17) (18) (19) (20) (21) (22) (23) (24) (25) (26) (27) (28) (29) (30) (31)

3. **Estrus cycle** is 18 to 24 Days long & the heat period is 14 hours

4. The gestation "pregnancy period" → 9 to 9.5 months (283 - 285 days)

5. Signs of estrus "estrogen hormone" اسبب في هارمون

Behavioral signs

Physical signs

- dec. Feed intake
- Urinate frequently يتبول كثير
- Bull Seeking behavior: يتنور على مكان التور
cow searches grassing beside bull وتأكل جنبه
- Homosexual activity: mounted by other cow
- cow may fail to take her customary place in the milking parlour
- Swelling & reddening in vulva
- Mucous discharges from vulva إفرازات مخاطية
- Dirty flanks when cow have been mounted by other cow

place in the milking parlour

Factors control Estrus Signs

Internal Factors

External Factors

① Genetic difference:

Swiss breed → least marked estrus

Black-Colored cattle → stronger signs

② Hormonal level

high estrogen hormone, signs of estrus become clear
 بزيادة العلامة أو فتح

① Nutritional factor

② Many Types of stress: affect milk production

↑ estrus ↓ Milk Production

③ Type of housing "Type of the floor"

Sandy or drith is better than concrete

④ Feet & leg problem: less mounting behavior activity
 خاص لدي مكتبة الفتح

Mennettullah Mahmoud

Detection of Estrous

① Herdman: Expert men who observe behavioral & physical signs 3 times a day
 Faults of this sys. 19% → are silent heat (ovulation with no signs) 15% → are false heat (signs of estrous with no ovulation)

② Bull: "Best detector of estrous in cows"

③ Vasectomized bull: bull with a surgical deviated penis
 ④ Freemartin heifer: Treatment with indelible anologous
 ⑤ Commercial detection device: glassic vial with red dye

Mounting

⑥ Tail head marking: marking the tail head with a chalk, paint or crayon
 ⑦ Colored chin ball: that's used ball on chin of bull that prints on female's got mounted
 ⑧ Pedometer: measure frequency of movement in cows, if it increases → cow is in estrous
 ⑨ Recently Trained dogs: Trained to detect estrous cow (80%-90% correct)
 ⑩ Progesterone Conc in milk: ↑ high during diestrus & inc further in pregnancy

⑪ Vaginal electrical Resistance: measure the electrical conductivity in Vaginal tissue & Uterine wall
 ⇒ Bull ←

① During Proestrous:

① Approaching & guarding ② Sniffing and Flehmen
 ③ Nudging ④ Mounting attempts till it stand

② During Estrous:

① Mounting ② Ejaculation: noted by marked generalized muscular contraction

((Proper time of Service or Insemination))

Cow normally ovulate about 12h after the end of heat period, if signs of estrous appear during morning the best time of service is the evening

البقرة ببصلا تبورض بعد 12 ميلة هي نهاية البقرة، لو العلامات ظهرت المبح أسمن وقت التفتح يكون بيل خاص لذي مكتبة الفتح

Synchronization of Estrous

كل الإحداثيات تكون في نفس الوقت

Using some hormones to bring all females of herd in estrus within a short time.

* Advantages of Synchronization -

- 1) Reducing the cost and effort for detection of estrus
- 2) Artificial insemination becomes easier
- 3) The calf crop becomes more uniform
- 4) Reducing the breeding and calving season

• توفير وقت وجهد الكشف عن الإحصيات

• إتقان العمل بسهولة

• الحصول على صغار في نفس الوقت

• التحكم في موسم التكاثر وتقليل الجوع

* Disadvantages of Synchronization -

- 1) Sometimes there is a lower conception rate

• أحياناً نسبة الحمل منخفضة

Artificial Insemination

- 1) Prevent of Venereal disease
- 2) Use of Superior bulls for much numbers of cows in the same herd or another
- 3) Reduces the cost of using much number of bulls within the farm
- 4) Improves the genetic character of the progeny produced in the farm

• تجنب الأمراض المنقولة

• استخدام ذكور ممتازة في كثير من الإحصيات

Diagnosis of Pregnancy in Cows

- 1- Rectal palpation: examination of ovaries & uterus & arteries during early & mid stages
- 2- Vaginal examination: Cervical glands secretions becomes more gelatinous & forming a plug that could be seen by speculum & lamp
- 3- Ultrasound Scanning: using rectal probe after 42 days from Semination
- 4- Laboratory Tests:
 - a) Physico-chemical test → detection of O-globulin in blood (100% at 180 days of pregnancy)
 - b) Barium chloride Test → adding few drops 3-6 drops of BaCl_2 to the urine, stays clear in case of pregnancy, turbid in case of non-pregnancy
 - c) Oxidation reduction test → by adding 0.6 ml of sodium benzoate to 3ml of cows urine in case of pregnant cow the urine doesn't return to its normal clearance
 - d) detection of Progesterone Conc → presence of high level in milk after 24 days indicates pregnancy

طريقة الجنس

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خاص لدي مكتبة الفتح

Signs of Pregnancy

يعني، ان يوم بدون عسل

- (a) Cessation of estrus (every 21 days)
- (b) The cow becomes more Tractable
- (c) abdomen becomes more enlarged
- (d) during later stage it drops down of hollows appears in both flanks

تتويعن الجنين

Managements of Pregnant Cow

- (1) Feeding: → increasing the level of nutrition,
→ last 2 months of pregnancy cows should be supplied with extra feed

Steaming up
رفع غذائي

Functions of Steaming up:

- fetus development
- build up the body reserve for coming lactation
- increase dairy milk yield
- lengthens the lactation of cows, which milked for short periods
- slightly increase butter fat percent
- prevent difficulties of parturition
Such as retention of the placenta and milk fever

- (2) Watering → drinking clean fresh water

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INK Tasters